

Joseph Walton

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Professional Summary

AI Engineer with over 10 years of experience delivering data-driven systems across government, intellectual property, financial services and defence. Specialises in evaluating, testing and improving LLM and agentic AI systems, with hands-on experience in evidence-based experimentation, red teaming, observability and failure analysis. Strong at building solutions, defining practical metrics, and translating technical findings into clear recommendations for stakeholders.

Technical Skills

- **Programming & Data:** Python, SQL, Pandas, Polars, Apache Spark
 - **AI / LLM Evaluation:** LLM evaluation, RAG evaluation, agentic AI testing, evaluation harnesses, benchmarking, red teaming, prompt experimentation, observability, failure analysis
 - **NLP & Generative AI:** HuggingFace, Azure AI Foundry, LangChain, LangGraph, RAG, vector search
 - **Machine Learning:** Scikit-learn, TensorFlow, PyTorch
 - **MLOps & Engineering:** MLflow, Azure ML Pipelines, Vertex AI Pipelines, CI/CD
 - **Cloud Platforms:** Microsoft Azure, Google Cloud Platform, AWS
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Professional Experience

IBM

Lead Data Scientist

June 2025 - Present

Contract role delivering AI solutions for central government.

- Rescued underperforming AI projects by introducing evidence-based evaluation, observability and failure analysis practices.
- Designed and delivered an agent-based data processing solution using Azure AI Foundry, Azure Functions and LangGraph, saving case workers two hours per day and approximately £2M annually.

Tech Stack: Python, Polars, Azure AI Foundry, Azure Functions, LangGraph

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

Developed and operationalised NLP solutions to improve trade mark application outcomes in a government environment.

- Developed generative AI applications using evaluation based development.
- Constructed evaluations to rigorously test refinements to scaffolding and prompt-based workflows.
- Conducted red-team experiments to evaluate generative AI robustness and identify improvements against adversarial attacks.
- Tuned vector search, through systematic evaluation.
- Used relevance metrics and experimental evidence to refine search and recommendation systems, improving relevance by 20%.

Tech Stack: Python, Azure OpenAI, HuggingFace, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

Led cross-functional delivery of production machine learning solutions using large-scale text and image data, including MLOps, NLP and computer vision systems.

Machine Learning Engineer

November 2018 - February 2020

Researched, evaluated and deployed machine learning models for intellectual property protection.

Tech Stack: Vertex Pipelines, HuggingFace, Pyspark

Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

Delivered national-scale government data integration projects with a focus on secure data handling and scalable data pipelines, including a secondment to Barclays to develop regional economic indicators from payments data and continue the winning ONS x Barclays Hackathon work.

Purple Secure Systems

Software Engineer

August 2014 - October 2016

Developed big data processing systems for government and defence clients.

Education

University of Bath

Master of Science in Applied Mathematics

2016 - 2017

Bachelor of Science in Physics

2010 - 2013

Honors & Awards

Winner of ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).